



Jeremías Likerman

RESEARCHER - CONICET · ASSOCIATE PROFESSOR - UNIVERSITY OF BUENOS AIRES

Instituto de Estudios Andinos Don Pablo Groeber (IDEAN)

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Present

Research topics: Current research focuses on the prediction of natural fracturing in geological structures using dynamic and static numerical modeling, the assessment of geothermal potential in volcanic and folded terrains through thermo-mechanical simulations, and the geodynamic modeling of subduction zones in active continental margins.

Professional Summary: Geoscientist with over 15 years of academic and field experience in structural geology, geomechanics, and geothermal modeling. Specialized in the application of numerical and analogue modeling techniques to assess fracture patterns, thermal evolution of lithosphere, and geothermal potential. Internationally active researcher with strong publication output, successful grant leadership, and extensive teaching and supervisory roles.

Academic Qualifications

Postdoctoral Fellowship

Argentina/Spain

CONICET & UPC

2015 - 2018

- Research Topic: Thermal evolution of the oceanic lithosphere. Supervisors: Dr. Cristallini (CONICET-UBA) and Dr. Zlotnik (UPC)

Stanford Visiting Scholar

USA

STANFORD UNIVERSITY

2013

- Courses taken: Structural Geology and MATLAB in Geosciences

PhD in Geological Sciences

Argentina

UNIVERSITY OF BUENOS AIRES

2010 - 2015

- Thesis: Prediction of fracturing on folded geological surfaces using numerical modeling. Supervisor: Dr. Cristallini.

Bachelor's in Geological Sciences

Argentina

UNIVERSITY OF BUENOS AIRES

2005 - 2010

- Thesis: Structure and geology in the Tres Cruces Compound Anticline, Jujuy, Argentina. Supervisor: Dr. Cristallini.

Research & Professional Experience

Researcher

Argentina

CONICET

2017 - present

- Focus: Fracture prediction, geodynamic modeling, geothermal potential modeling

Associate Professor

Argentina

DEPARTMENT OF GEOLOGICAL SCIENCES, UNIVERSITY OF BUENOS AIRES

2013 - present

- Courses: Introduction to Geology, Geological Surveying, Geostatistics.

Shale Reservoir Geology

Argentina

COURSE INSTRUCTOR

2025

- Instructor of the professional development course focused on the geological characterization of shale reservoirs

Advanced Researcher Field Trip

Argentina

FIELD INSTRUCTOR

2025

- Organizer and instructor of the field trip: Perspectives on Vaca Muerta — subsurface attributes analyzed from the surface

Visiting Researcher

Spain

GEO3BCN-CSIC

2024-2025

- Project: LINGLOBAL – Geothermal potential estimation using numerical models

Visiting Researcher

BARCELONATECH (UPC)

- Lab: Numerical Computation

Spain

2015, 2017, 2019, 2022

Scholarships & Awards

2025	SZNet 2025 Chile Field Trip	National Science Foundation	SZ4D
2022	Ibero-American Postgraduate University Association	Academic Mobility Scholarships between Associated Institutions	AUIP
2016	National Scientific and Technical Research Council	International Research Stay Grant	CONICET
2015	National Scientific and Technical Research Council	Postdoctoral Fellowship	CONICET
2013	Stanford University	Fulbright - Bunge & Born Foundation	PhD Scholarship
2013	National Scientific and Technical Research Council	Doctoral Scholarship Type II	CONICET
2010	National Scientific and Technical Research Council	Doctoral Scholarship Type I	CONICET

Research Grants

2025	Geothermal potential of volcanic zones (PI).	– AECT-2025-1-0009	Spain
2025	Structural controls on volcanic systems: magmatic intrusions and the development of damage zones associated with hydrocarbon generation (co-PI).	– Fundación Williams	Argentina
2024	Estimation of geothermal potential in volcanic zones using thermomechanical numerical models (PI).	– LINCGL2024	Spain
2022	Modelling the geodynamics of subduction zones and mantle plumes (PI).	– AECT-2024-2-0027	Spain
2023	Geodynamics of the Adria microplate, from mantle to surface (co-PI).	– Geo3BCN	Spain
2022	Geodynamic modelling of subduction zones (PI).	– AECT-2022-3-0023	Spain
2020	Study of landscape dynamics and geological risks associated with dam placement in mountainous areas (co-PI).	– PICT-2019-04011	Argentina
2020	Analog and numerical modelling: contributions to solving tectonic and structural problems (co-PI).	– PICT-2019-00997	Argentina
2018	Natural fracturing and first-order structures: predictive models applied to Cerro Domuyo (co-PI).	– UBACyT	Argentina
2017	Geological evolution of the Andes and its economic and environmental impact (co-PI).	– PUE 22920160100051	Argentina
2016	Numerical modelling of the thermal evolution of the oceanic lithosphere (PI).	– PICT 2015-1226	Argentina
2016	Application of analog and numerical models to tectonic-structural problems (co-PI).	– PICT-2016-1407	Argentina

Supervision

POSTDOCTORAL SUPERVISION

Plotek, Berenice

FAULT PROPAGATION FOLDS: MECHANICAL APPROACH THROUGH FINITE ELEMENT NUMERICAL MODELLING

- **Advisors:** Cristallini, E. and Likerman, J.

Argentina

2024

PHD THESIS SUPERVISION

Correa Luna, Clara

NATURAL FRACTURING IN THE VACA MUERTA FORMATION. CHARACTERIZATION AND MODELLING

- **Advisors:** Yagupsky, D. and Likerman, J.

Argentina

2023

BACHELOR'S THESIS SUPERVISION

Chandía Sepúlveda, Alexander Ignacio

TECTONIC INVERSION OF A METAMORPHIC COMPLEX IN THE DESEADO MASSIF: STRUCTURAL INSIGHTS FROM ANALOG MODELLING

- **Advisors:** Likerman, J. and Navarrete, C.

Argentina

In progress

Celis Santisteban, Celso Angel

Argentina

STRUCTURAL EVOLUTION OF AN INVERTED METAMORPHIC COMPLEX IN THE DESEADO MASSIF: INSIGHTS FROM
NUMERICAL MODELLING

In progress

- **Advisors:** Likerman, J. and Navarrete, C.

De Rosa, Tomás

Argentina

QUATERNARY DEFORMATION IN THE FIAMBALÁ VALLEY, CATAMARCA

In progress

- **Advisors:** Likerman, J. and Sagripanti, L.

Acosta, Luana

Argentina

GEOMORPHOLOGY OF THE GRANDE RIVER VALLEY BETWEEN BARDAS BLANCAS AND PORTEZUELO DEL VIENTO,
MENDOZA

2023

- **Advisors:** Winocur, D. and Likerman, J.

Mazzini, Martín

Argentina

GEOLOGY OF THE EASTERN SLOPE OF CERRO PERITO MORENO, EL BOLSÓN, RÍO NEGRO PROVINCE

2022

- **Advisors:** Likerman, J. and Tobal, J.

Relañez, Gastón

Argentina

STRUCTURE AND NEOTECTONICS OF THE TABLADITAS AREA, JUJUY PROVINCE, ARGENTINA

2019

- **Advisors:** Yagupsky, D. and Likerman, J.

Ventisky, Esteban

Argentina

GEOLOGY, STRUCTURE, AND FRACTURING OF THE LAJAS FORMATION IN THE COVUNCO ANTICLINE, PORTADA
COVUNCO, NEUQUÉN

2017

- **Advisors:** Likerman, J. and Tobal, J.

Correa Luna, Clara

Argentina

STRUCTURAL GEOLOGY AND FRACTURING IN THE TRES CRUCES BASIN, JUJUY PROVINCE

2015

- **Advisors:** Yagupsky, D. and Likerman, J.

Publications

Peer-Reviewed Publications

Impact of large-scale structures on fracture network connectivity: Insights into the Vaca Muerta unconventional play, Neuquén basin, Argentina

Correa-Luna, C., Yagupsky, D. L., **Likerman, J.**, Barcelona, H.

Tectonophysics (2025) p. 230640. Elsevier, 2025

Slab underthrusting is the primary control on flat slab size

Gianni, G. M., Gallo, L. C., **Likerman, J.**, Echaurren, A., Gianni, C. R., Faccena, C.

Science Advances (2025). In Press, 2025

Geomechanical modeling of fault-propagation folds: A comparative analysis of finite-element and the trishear kinematic model

Plotek, B., **Likerman, J.**, Cristallini, E.

Journal of Structural Geology (2024) p. 105064. Elsevier, 2024

Ghost-arc geochemical anomaly at a spreading ridge caused by supersized flat subduction

Gianni, G. M., **Likerman, J.**, Navarrete, C. R., Gianni, C. R., Zlotnik, S.

Nature communications 14.1 (2023) p. 2083. Nature Publishing Group UK London, 2023

Massive Jurassic slab break-off revealed by a multidisciplinary reappraisal of the Chon Aike silicic large igneous province

Navarrete, C., Gianni, G., Tassara, S., Zaffarana, C., **Likerman, J.**, Márquez, M., Wostbrock, J., Planavsky, N., Tardani, D., Frasette, M. P.

Earth-Science Reviews (2023) p. 104651. Elsevier, 2023

Natural fracture characterization of the Los Catutos Member (Vaca Muerta Formation) in the southern sector of the Sierra de la Vaca Muerta, Neuquén Basin, Argentina

Correa-Luna, C., Yagupsky, D. L., **Likerman, J.**, Barcelona, H.

Journal of South American Earth Sciences 120 (2022) p. 104081. Elsevier, 2022

Kinematics of fault-propagation folding: Analysis of velocity fields in numerical modeling simulations

Plotek, B., Heckenbach, E., Brune, S., Cristallini, E., **Likerman, J.**

Journal of Structural Geology 162 (2022) p. 104703. Elsevier, 2022

The influence of climate on the dynamics of mountain building within the Northern Patagonian Andes

García Morabito, E., Beltrán-Triviño, A., Terrizzano, C. M., Bechis, F., **Likerman, J.**, Von Quadt, A., Ramos, V. A.

Tectonics 40.2 (2021) e2020TC006374. Wiley Online Library, 2021

The effects of small-scale convection in the shallow lithosphere of the North Atlantic

Likerman, J., Zlotnik, S., Li, C.-F.

Geophysical Journal International 227.3 (2021) pp. 1512–1522. Oxford University Press, 2021

Fracture network analysis of Yacoraite Formation in the Tres Cruces sub-basin, northwestern Argentina

Luna, C. C., Yagupsky, D. L., **Likerman, J.**

Journal of South American Earth Sciences (2019) p. 102446. Elsevier, 2019

Cenozoic intraplate tectonics in Central Patagonia: Record of main Andean phases in a weak upper plate

Gianni, G. M., Echaurren, A., Folguera, A., **Likerman, J.**, Encinas, A., García, H. P. A., Dal Molin, C., Valencia, V. A.

Tectonophysics 721 (2017) pp. 151–166. Elsevier, 2017

Closure type effects on the structural pattern of an inverted extensional basin of variable width: Results from analogue models

Jara, P., **Likerman, J.**, Charrier, R., Herrera, S., Pinto, L., Villarroel, M., Winocur, D.

Journal of South American Earth Sciences (2017). Elsevier, 2017

Climatic and Tectonic forcing on alluvial fans in the Southern Central Andes

Terrizzano, C. M., Morabito, E. G., Christl, M., **Likerman, J.**, Tobal, J., Yamin, M., Zech, R.

Quaternary science reviews 172 (2017) pp. 131–141. Elsevier, 2017

Structural and statistical characterization of joints and multi-scale faults in an alternating sandstone and shale turbidite sequence at the Santa Susana Field Laboratory: Implications for their effects on ground-water flow and contaminant transport

Cilona, A., Aydin, A., **Likerman, J.**, Parker, B., Cherry, J.

Journal of Structural Geology 85 (2016) pp. 95–114. Elsevier, 2016

Middle to late Miocene extensional collapse of the North Patagonian Andes (41° 30'–42° S)

Tobal, J. E., Folguera, A., **Likerman, J.**, Naipauer, M., Sellés, D., Boedo, F. L., Ramos, V. A., Gimenez, M.

Tectonophysics 657 (2015) pp. 155–171. Elsevier, 2015

Geodynamic context for the deposition of coarse-grained deep-water axial channel systems in the Patagonian Andes

Ghiglione, M. C., **Likerman, J.**, Barberón, V., Giambiagi, L. B., Aguirre-Urreta, B., Suarez, F.

Basin Research 26.6 (2014) pp. 726–745. Wiley Online Library, 2014

Role of basin width variation in tectonic inversion: insight from analogue modelling and implications for the tectonic inversion of the Abanico Basin, 32°–34° S, Central Andes

Jara, P., **Likerman, J.**, Winocur, D., Ghiglione, M. C., Cristallini, E. O., Pinto, L., Charrier, R.

Geological Society, London, Special Publications 399 (2014) SP399–7. Geological Society of London, 2014

Along-strike structural variations in the Southern Patagonian Andes: Insights from physical modeling

Likerman, J., Burlando, J. F., Cristallini, E. O., Ghiglione, M. C.

Tectonophysics 590 (2013) pp. 106–120. Elsevier, 2013

Skills

Programming

LaTeX, Python, HTML5, R, MatLab

Modeling Tools

- **Numerical methods:** Underworld2, Golem
- **Analogue modelling:** setup, material properties, PIV, post-processing

Fieldwork

- Extensive fieldwork experience (over 15 years)
- broad knowledge of the Andes
- Planning and logistics of field expeditions
- Team leadership

Languages

- Spanish (native)
- English (fluent C2)
- Italian (A2)
- Catalan (A2)